Stage 1 Report — Macroeconomic Forecasting with Large Language Models

Paper reproduced: Carriero, Pettenuzzo & Shekhar (2026), *"Macroeconomic Forecasting with Large Language Models,"* arXiv:2407.00890v5.

1. The Paper, in Brief

Problem. Can Time Series Foundation Models (TSFMs) — pretrained transformer models for time series, the "LLMs of forecasting" — beat traditional econometric methods at forecasting macroeconomic variables?

Data. FRED-MD: 120 monthly US macro series (output, prices, labor, housing, money/credit, rates/FX, stock market), Jan 1959–Dec 2022, each series transformed to near-stationarity via McCracken & Ng (2016)'s standard codes (log-differences, first differences, etc.).

Models compared.

- *Econometric:* AR(1) (mandatory benchmark), BVAR with a Minnesota-tradition Normal-Inverse-Wishart prior (p=6 lags, hyperparameters tuned by maximising the marginal likelihood), a PCA-factor-augmented autoregression, NNAR, and BART.
- *TSFMs (all zero-shot):* Chronos (Amazon), Moirai (Salesforce), TimesFM (Google), Lag-Llama, TTM (IBM), TimeGPT (Nixtla) — ranging 1M–311M parameters.
- *Hybrid:* fit BVAR then take its in-sample residuals, forecast those residuals zero-shot with a TSFM, and add the two forecasts together (their "TSFM-BVAR").

Method. Recursive expanding-window forecasting: train on Jan 1960–Dec 1984, forecast Jan 1985 (h=1..12), roll forward one month, repeat through Dec 2022. Accuracy is measured as RMSFE relative to the AR(1) benchmark (their Eq. 11):

```
RMSFE_ijh = sqrt( Sum e^2_{i,j,tau+h} / Sum e^2_{AR1,j,tau+h} )
```

Values below 1 mean model *i* beats AR(1) for series *j* at horizon *h*.

Headline findings.

1. Only 3 of 6 TSFMs (Chronos, TimesFM, Moirai-large) are consistently competitive with AR(1); TTM, Lag-Llama, TimeGPT lag behind.
2. Even the good TSFMs are only roughly on par with BVAR/Factor models, not clearly better.
3. BVAR and TSFM forecast errors are nearly *orthogonal* at h=1 (correlation ~0.07–0.5) — they're wrong about different series — which is exactly why combining them (the hybrid) helps: it substantially trims the worst-case errors of the TSFM.
4. TSFMs handle the COVID-19 shock better than econometric models (with the caveat that their training data may include the pandemic period).

2. What We Actually Built (Stage 1)

Full pipeline, all runnable from `/Users/nilesh/LLM`:

Script	Function
`scripts/prep_data.py`	Loads FRED-MD CSV, selects 60-series subset, applies the 7 McCracken-Ng transform codes, saves `data/fredmd_clean.parquet`
`scripts/econ_models.py`	AR(1), Minnesota-prior BVAR, PCA-factor-augmented AR — shared `_bvar_fit()` used by both the forecast function and the hybrid's residual extraction
`scripts/run_forecast_loop.py`	Recursive expanding-window loop + `rmsfe_table()` (paper's Eq. 11)
`scripts/run_tsfm_chronos.py`	Amazon Chronos (bolt-small, zero-shot)
`scripts/run_tsfm_timesfm.py`	Google TimesFM 2.5 (200M, zero-shot)
`scripts/run_tsfm_moirai.py`	Salesforce Moirai (base, 91M, zero-shot)
`scripts/run_tsfm_ttm.py`	IBM Tiny Time Mixer (5M, zero-shot)
`scripts/run_tsfm_lagllama.py`	Lag-Llama (25M, zero-shot)
`scripts/run_tsfm_timegpt.py`	Nixtla TimeGPT via API (batched: 1 call/origin covering all 60 series, not 1 call/series)
`scripts/run_hybrid.py`	TSFM-BVAR hybrid: BVAR forecast + TimesFM-modelled BVAR residuals
`scripts/merge_tsfm_errors.py`	Folds any model's raw forecasts into the shared error table
`scripts/make_plots.py`	Boxplots mirroring the paper's Figure 2/3

Environments: `venv_core` (Python 3.14 — torch, chronos-forecasting, timesfm, sklearn, statsmodels), `venv_moirai` and `venv_lagllama` (Python 3.11, forced by `uni2ts`/`granite-tsfm` pinning scipy/torch versions). All six TSFMs and both econometric benchmarks actually ran to completion on real data; nothing here is simulated or hand-waved.

Two real implementation bugs were found and fixed during development: 1. **BVAR dummy-prior bug:** a lag-decay multiplier was accidentally a no-op (`lag if lag==1 else 1` always evaluates to 1), so the Minnesota prior wasn't actually shrinking higher lags. Fixed to scale properly as `lag * sigma_i / lambda` (paper's Eq. B.1). Also switched the prior mean from 1 (random-walk prior, correct for *levels*) to 0, since FRED-MD series are already transformed to near-stationarity. 2. **Factor model instability:** the original implementation iterated a 1-step model forward 12 times to get an h=12 forecast; any autoregressive coefficient slightly above 1 explodes exponentially under iteration, producing RMSFE outliers as large as 246x. Replaced with the paper's actual specification (Eq. 6-7): a **direct** regression of y_t on y_{t-h} and f_{t-h}, one regression per horizon, no iteration.

3. Our Approach versus the Original Paper

The project follows the paper's logic: same dataset, same relative-RMSFE metric, same family of benchmarks and the same BVAR-plus-TSFM hybrid. It differs in scale and in several design choices, summarised in Table 1. Notably, we add Chronos-Bolt and newer TimesFM versions, while the paper's additional nonlinear benchmarks (NNAR, BART), significance tests and fine-tuning experiments are not yet replicated.

Table 1. Comparison of the original study and this work.

Aspect	Original paper (Carriero et al.)	This work
Data	FRED-MD, 120 monthly series, 1959–2022	FRED-MD, 60 balanced series, 1959–2022
Evaluation	Jan 1985 – Dec 2019 (main), 2020–2022 separately; monthly origins; h = 1, ..., 12	2010 Q1 – 2022 Q4; 48 quarterly origins; h = 1, 3, 6, 12
Benchmarks	AR(1), BVAR, Factor model, NNAR, BART	AR(1), BVAR, Factor-AR
BVAR prior	Minnesota + sum-of-coefficients + single-unit-root; p = 6; hyper-parameters by marginal likelihood	Minnesota prior; p = 6; lambda = 0.2 fixed
TSFMs	MOIRAI (base, large), TimesFM, TTM, TimeGPT, Lag-Llama	TimesFM 2.0/2.5, TimeGPT, Chronos-Bolt, MOIRAI, TTM, Lag-Llama
Hybrid	BVAR + MOIRAI-large and BVAR + TimesFM	BVAR + TimesFM
Extra analyses	Diebold-Mariano tests, COVID split, data-leakage test, persistence split, fine-tuning	Not yet implemented

4. Results and Comparison with the Paper

Table 2 reports the results of this work and Table 3 those of the original paper, each covering the single models and the hybrid models. All entries are the median RMSFE relative to AR(1) across series (values below 1, in **bold**, beat AR(1)). Because the two studies use different series sets, evaluation windows and model versions, the comparison is about patterns rather than exact decimals.

Table 2. Results of this work: median relative RMSFE (60 FRED-MD series, 2010 Q1 – 2022 Q4, 48 quarterly origins).

Model	h = 1	h = 3	h = 6	h = 12
Single models				
AR(1) (benchmark)	1.000	1.000	1.000	1.000
BVAR (Minnesota)	1.332	1.262	1.167	1.108
Factor-AR	1.507	1.192	1.089	1.029
TimesFM 2.0/2.5	**0.983**	**0.995**	**0.997**	1.000
TimeGPT	**0.987**	**0.997**	**0.994**	**0.999**
Chronos-Bolt	**0.996**	**0.997**	**0.998**	1.002
MOIRAI	1.031	1.004	1.003	1.013

Model	h = 1	h = 3	h = 6	h = 12
TTM	1.117	1.040	1.019	1.007
Lag-Llama	1.043	1.022	1.020	1.014
Hybrid model				
BVAR + TimesFM	1.331	1.250	1.168	1.111

Chronos-Bolt was evaluated only in this work. The hybrid combines the BVAR with TimesFM residual forecasts.

Table 3. Results of the original paper: median relative RMSFE (120 FRED-MD series, Jan 1985 – Dec 2019, monthly origins).

Model	h = 1	h = 3	h = 6	h = 12
Single models – econometric				
AR(1) (benchmark)	1.000	1.000	1.000	1.000
BVAR	**0.983**	**0.970**	**0.991**	**0.999**
Factor model	**0.965**	**0.980**	**0.992**	**0.999**
NNAR	**0.988**	**0.992**	**0.999**	1.000
BART	1.017	1.024	1.011	1.023
Single models – zero-shot TSLMs				
MOIRAI-base	1.008	**0.973**	**0.990**	**0.999**
MOIRAI-large	**0.999**	**0.978**	**0.991**	1.001
TimesFM	1.014	**0.980**	**0.990**	1.001
TimeGPT	1.077	1.044	1.025	1.063
TTM-Enhanced	1.044	**0.993**	**0.995**	1.002
Lag-Llama	1.015	**0.997**	1.002	1.009
Hybrid models				
BVAR + MOIRAI-large	**0.973**	**0.960**	**0.983**	**0.998**
BVAR + TimesFM	**0.980**	**0.956**	**0.984**	**0.999**

Source: arXiv:2407.00890v4, Tables 2-4. The repository cites a later revision, so figures may differ slightly. In Table 3 and in the text, MOIRAI refers to MOIRAI-large, TTM to TTM-Enhanced and the hybrid to the BVAR + TimesFM version.

4.1 Pre-COVID Results (2010–2019 Origins)

Clean comparison mirroring the paper's own choice to exclude the pandemic period from its main figure:

Model	h = 1	h = 3	h = 6	h = 12
BVAR	1.000	1.055	1.007	0.997

Model	h = 1	h = 3	h = 6	h = 12
Factor-AR	**0.981**	**0.990**	**0.994**	1.002
TimesFM	**0.965**	**0.981**	**0.984**	0.997
Chronos	**0.991**	**0.987**	**0.986**	0.999
TimeGPT	**0.980**	**0.985**	**0.988**	0.998
MOIRAI	1.032	1.025	1.014	1.005
TTM	1.072	1.011	1.000	1.000
Lag-Llama	1.079	1.093	1.068	1.014
Hybrid (BVAR + TimesFM)	**0.989**	1.050	1.008	0.998

4.2 COVID Stress Test — Full vs. COVID-Only Window

Full window (2010–2022) vs. COVID-only subset (2019-06 onward) — shows model fragility to the regime break:

Model	Full h = 1	COVID-only h = 1	Full h = 12	COVID-only h = 12
BVAR	1.332	1.564	1.108	1.873
Factor-AR	1.507	1.897	1.029	1.096
TimesFM	**0.983**	**0.973**	1.000	1.011
Chronos	**0.996**	1.022	1.002	1.000
TimeGPT	**0.987**	**0.981**	**0.999**	1.002
Hybrid	1.331	1.578	1.111	1.880

Full precision numbers and std/min/max in `output/rmsfe_summary.csv`; boxplots in `output/rmsfe_boxplots.png`.

4.3 Interpretation

TSFM ranking replicates the paper closely. Our best-to-worst ordering — TimesFM ~ Chronos ~ TimeGPT (all beat AR(1) at every horizon) > MOIRAI (roughly at par) > TTM ~ Lag-Llama (worse than AR(1)) — matches the paper's own ranking closely with one clear divergence, discussed below.

BVAR/Factor models beat AR(1) pre-COVID, as the paper found, but their performance is not stable across regimes — the striking finding here is how badly they break down once COVID enters the sample (RMSFE up to 1.87 vs AR(1) at h=12), while every TSFM stays within a few percent of 1.0 throughout. This is a real-data confirmation of the paper's own conclusion (section 8): *"their [TSFMs'] relatively stronger performance in the post-Covid-19 period suggests an advantage in handling structural breaks."*

TimeGPT is our biggest divergence from the paper. The paper found it one of the two weakest TSFMs (median 1.03–1.08); here it's one of the best (0.98–0.99). Most likely explanation: TimeGPT is the only TSFM served via a live, continuously-updated hosted API — we are almost certainly querying a materially different (newer) model than the one benchmarked in the paper's 2024 evaluation.

The hybrid partially replicates the paper's benefit, but not fully. It improves on plain BVAR at h=1 (1.000 to 0.989) and clearly reduces the tail risk. The COVID blowup is inherited rather than fixed (Hybrid tracks BVAR's 1.56–1.87 blowup) because the hybrid's forecast is anchored to BVAR's own point forecast — the residual correction only fixes the linear, learnable part of BVAR's error, not a wholesale level shift. Our simplified BVAR (no stochastic volatility, no outlier handling) is comparatively weaker than the paper's fully-tuned version, capping the hybrid's ceiling below standalone TimesFM.

5. Conclusion

The core qualitative finding of the paper — TSFMs are competitive with but not clearly superior to well-specified econometric models, and the two are complementary enough that combining them can help — replicates at our reduced scale. The TSFM performance ranking matches closely; the divergence on TimeGPT and the more limited hybrid gain are both explainable by concrete implementation differences (hosted API drift, a less richly-specified BVAR prior) rather than methodological errors, and both are documented above per Stage 1's requirement to discuss and justify any differences from the original paper.